

Assignment 1 — Univariate and Multivariate Statistics

1. Reading the Shape of a Sample

Predict

For `problem1.csv`, I calculated the first four sample moments:

- Mean = **0.001104**
- Variance = **0.00009625**
- Skewness = **-0.6698**
- Excess kurtosis = **2.3576**

The negative skewness means the distribution has a longer left tail. The positive excess kurtosis means the tails are heavier than those of a Normal distribution.

Based only on these two moments, I would rule out the Normal distribution because a Normal should have zero skewness and zero excess kurtosis. I would also rule out the Student's t distribution because it is symmetric, so its skewness should also be zero.

A lognormal distribution is not a good choice either because it is positively skewed and only takes positive values.

Among the distributions discussed in class, the Normal Inverse Gaussian distribution seems the most plausible because it can allow both skewness and heavy tails.

Fit and Reconcile

I fit a Normal distribution using the sample mean and variance. The fitted standard deviation is about **0.00981**, and the 1% quantile is

$$q_{0.01} = -0.02172.$$

There are **26 observations** below this value.

Since there are 1,000 observations, a correctly fitted Normal would predict about

$$1000 * 0.01 = 10$$

observations below its 1% quantile.

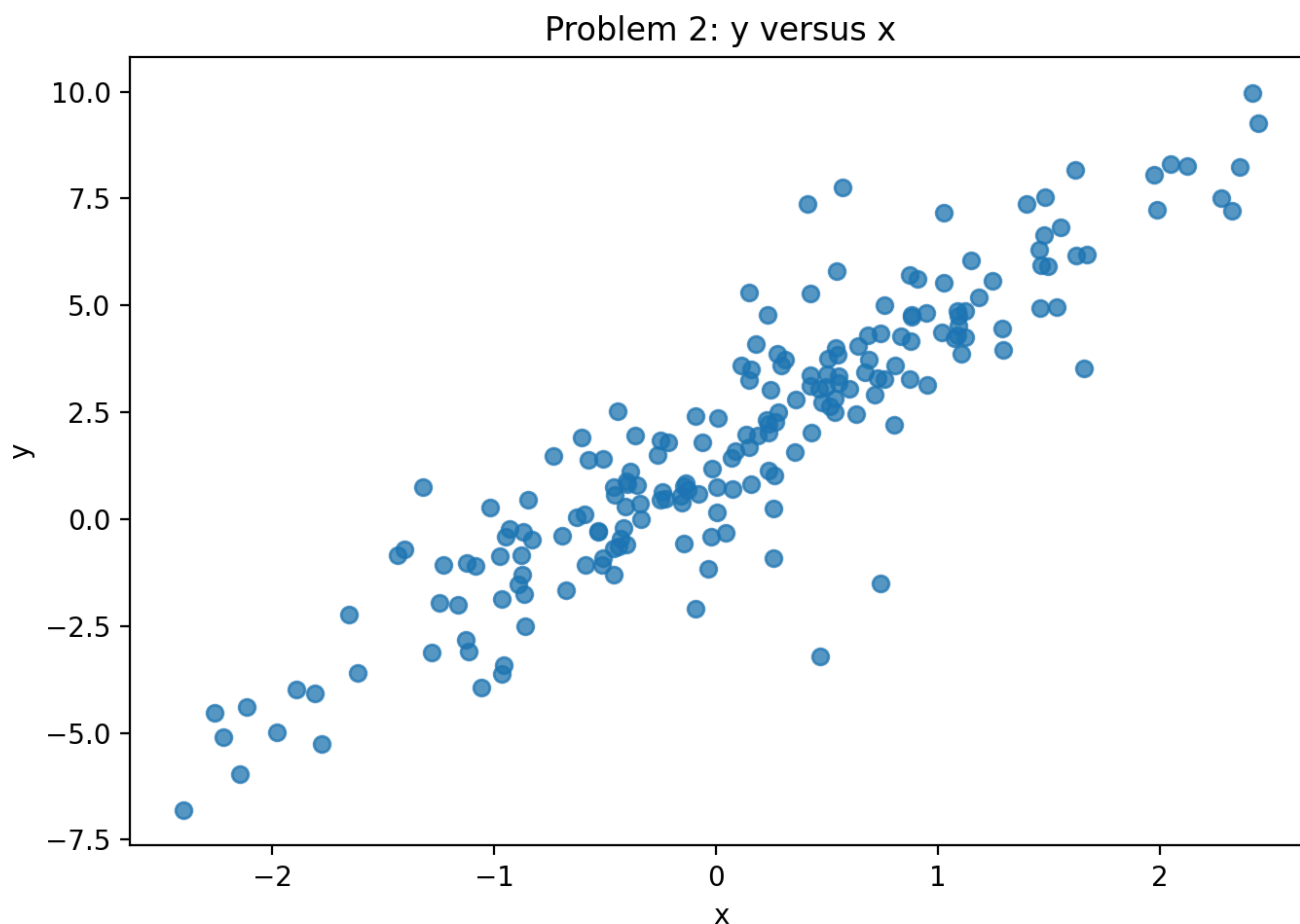
So the comparison is:

- Actual: **26**
- Expected under Normal: **10**

This is consistent with what I expected from the moments. The Normal model does not capture the left tail well. It underestimates the probability of large negative outcomes, which would be especially important in a risk management setting.

2. A Regression Whose Errors Are Not Normal

Predict



The scatterplot shows a very clear linear relationship between x and y . However, there are also several observations that are much farther away from the main line than the others.

The errors look roughly symmetric, but they appear to have heavy tails. Because of this, I expect a Student's t error distribution to fit better than a Normal error distribution.

The OLS assumption being violated is the assumption that the errors are normally distributed.

I do not expect this violation to change the slope very much because the basic linear relationship is still very clear and the large residuals do not appear to occur only at high or low values of x . However, I expect the Normal model to have trouble describing the uncertainty in the errors, especially in the tails.

Fit

The OLS estimates are:

$$\hat{\alpha} = 1.5607$$

$$\hat{\beta} = 2.9908.$$

The standard errors are approximately:

$$SE(\hat{\alpha}) = 0.0963$$

$$SE(\hat{\beta}) = 0.0966.$$

For the Normal-error maximum likelihood model, I get:

$$\hat{\alpha} = 1.5607$$

$$\hat{\beta} = 2.9908$$

$$\hat{\sigma} = 1.3446.$$

Its AICc is:

$$692.145.$$

For the Student's t -error model, I get:

$$\hat{\alpha} = 1.5480$$

$$\hat{\beta} = 3.0192$$

$$\hat{\sigma} = 0.9355$$

$$\hat{\nu} = 3.6322.$$

Its AICc is:

$$665.327.$$

Since a lower AICc is better, the Student's t model is strongly preferred.

Reconcile

The estimated slopes are:

- OLS: **2.9908**
- Normal MLE: **2.9908**
- Student's t MLE: **3.0192**

The slopes are very close, so the violation of Normality did not change the estimated relationship between x and y very much.

The main difference is instead in the error distribution. The Normal model has to use a relatively large standard deviation to account for the extreme residuals. The Student's t model can keep a tighter center while also allowing much heavier tails.

The upper error quantiles are:

Quantile	Normal	Student's t
95%	2.212	2.054
99.5%	3.464	4.620

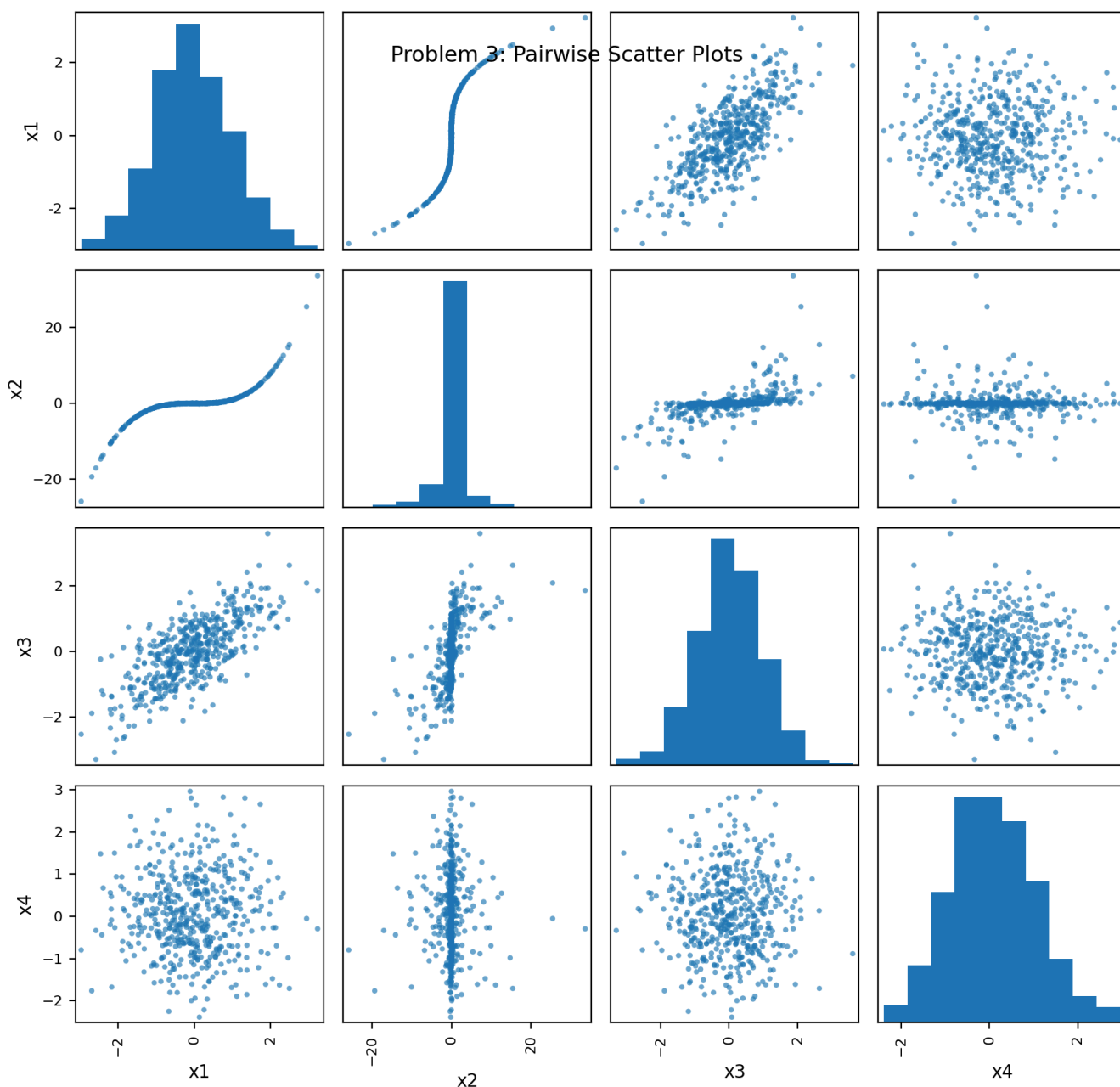
At 95%, the Normal is slightly wider. At 99.5%, the Student's t is much wider.

This happens because the Normal spreads probability more evenly by increasing its overall scale, while the Student's t concentrates more probability near the center but also puts much more probability in the extreme tails.

If I were setting a capital buffer, I would prefer the Student's t model because capital buffers are mainly intended to protect against rare and severe losses. Therefore, the far tail is more important than the middle of the distribution.

3. Pearson Against Spearman

Predict



From the pair plots, I expect x_1 and x_2 to have the largest difference between Pearson and Spearman correlation.

The relationship looks strongly increasing but nonlinear. As x_1 gets larger, x_2 also gets larger almost all the time, but not at a constant rate.

Since Spearman only cares about the ranking and monotonic relationship, I expect it to be very high. Pearson measures linear association, so I expect it to be lower.

For pairs involving x_4 , I expect Pearson and Spearman to agree that there is almost no relationship.

Fit and Reconcile

The Pearson correlation between x_1 and x_2 is approximately

$$0.7991$$

while the Spearman correlation is approximately

$$0.9719.$$

The gap is therefore

$$|0.7991 - 0.9719| = 0.1728$$

which is the largest among all pairs.

This matches my prediction.

The reason for the gap is that the relationship is strongly monotonic but nonlinear. The observations are ordered almost perfectly as x_1 increases, so Spearman is very high. However, the relationship is curved rather than linear, so Pearson is lower.

For the question "do these two variables consistently move in the same direction?", Spearman is the better description.

Pearson is answering a different question: "how well can this relationship be represented by a straight line?"

4. Conditional Distributions

Predict

The sample covariance matrix is

$$\Sigma = \begin{pmatrix} 1.09988 & 1.69690 \\ 1.69690 & 4.10475 \end{pmatrix}.$$

Here,

$$\Sigma_{11} = \text{Var}(x_1) = 1.09988$$

$$\Sigma_{22} = \text{Var}(x_2) = 4.10475$$

and

$$\Sigma_{12} = \Sigma_{21} = \text{Cov}(x_1, x_2) = 1.69690.$$

The conditional variance of x_2 given x_1 is

$$Var(x_2|x_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}.$$

Using the sample values,

$$Var(x_2|x_1) = 1.48675.$$

The remaining fraction of variance after observing x_1 is

$$\frac{1.48675}{4.10475} = 0.3622.$$

So knowing x_1 leaves about **36.2%** of the original variance in x_2 , which means uncertainty is reduced by about **63.8%** in variance terms.

Under the multivariate Normal assumption, this factor does not depend on the actual value of x_1 . The conditional variance formula only contains elements from the covariance matrix and does not contain the observed value of x_1 .

Fit and Reconcile

The conditional mean is

$$E[x_2|x_1] = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1).$$

The sample means are:

$$\mu_1 = -0.01671$$

$$\mu_2 = 0.04529.$$

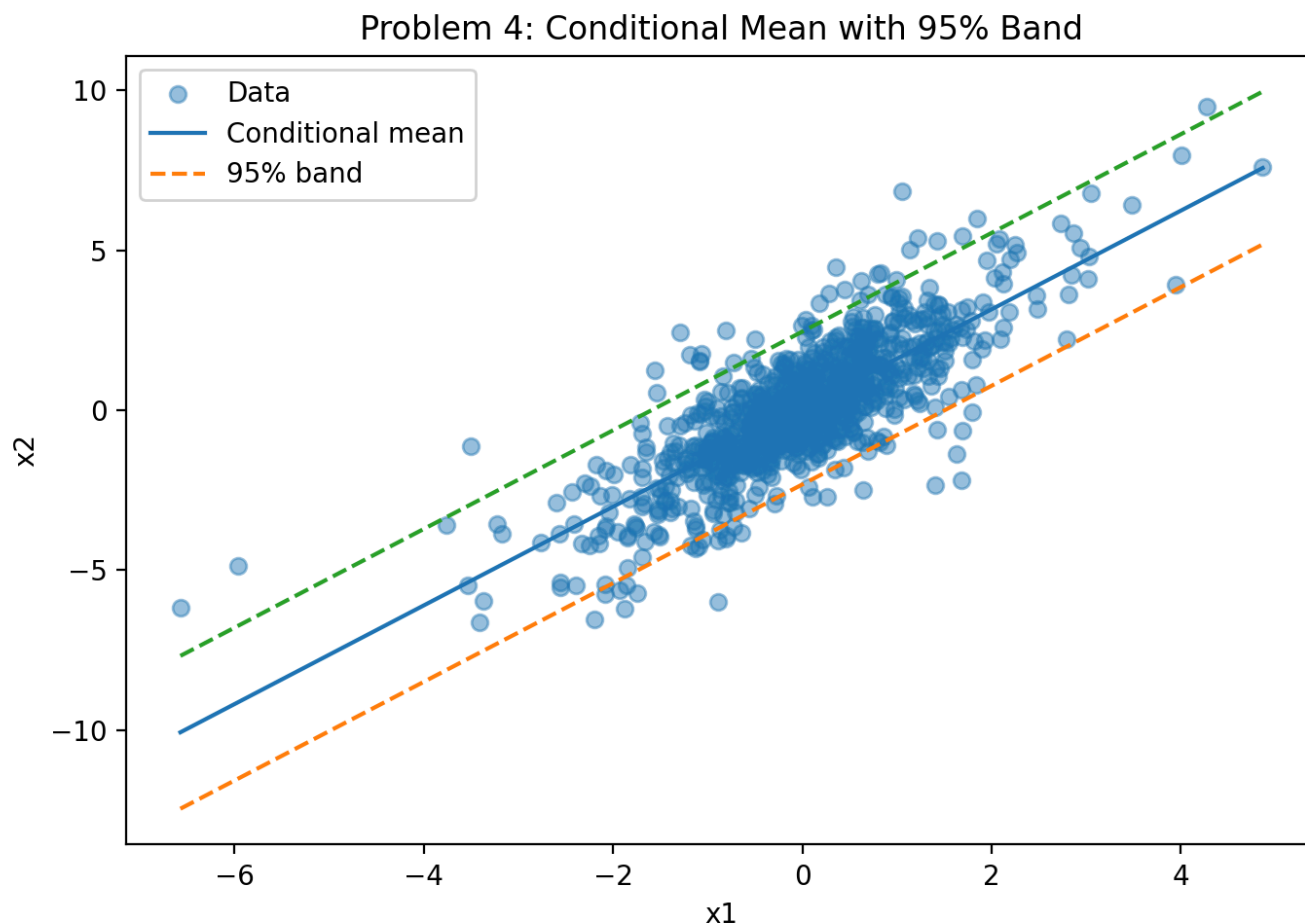
The coefficient on x_1 is

$$\frac{\Sigma_{21}}{\Sigma_{11}} = 1.54281.$$

This is also the simple OLS slope from regressing x_2 on x_1 .

So the conditional expectation can be written as

$$E[x_2|x_1] = 0.07107 + 1.54281x_1.$$



Using the constant conditional variance, the 95% band has a constant half-width of about
2.390.

Overall, **93.5%** of observations fall inside the nominal 95% band.

When I split the observations according to how far x_1 is from its mean, I get:

Distance from mean	Coverage
Within 1 SD	95.65%
Between 1 and 2 SD	85.79%
Beyond 2 SD	90.20%

The coverage is not constant across the three groups.

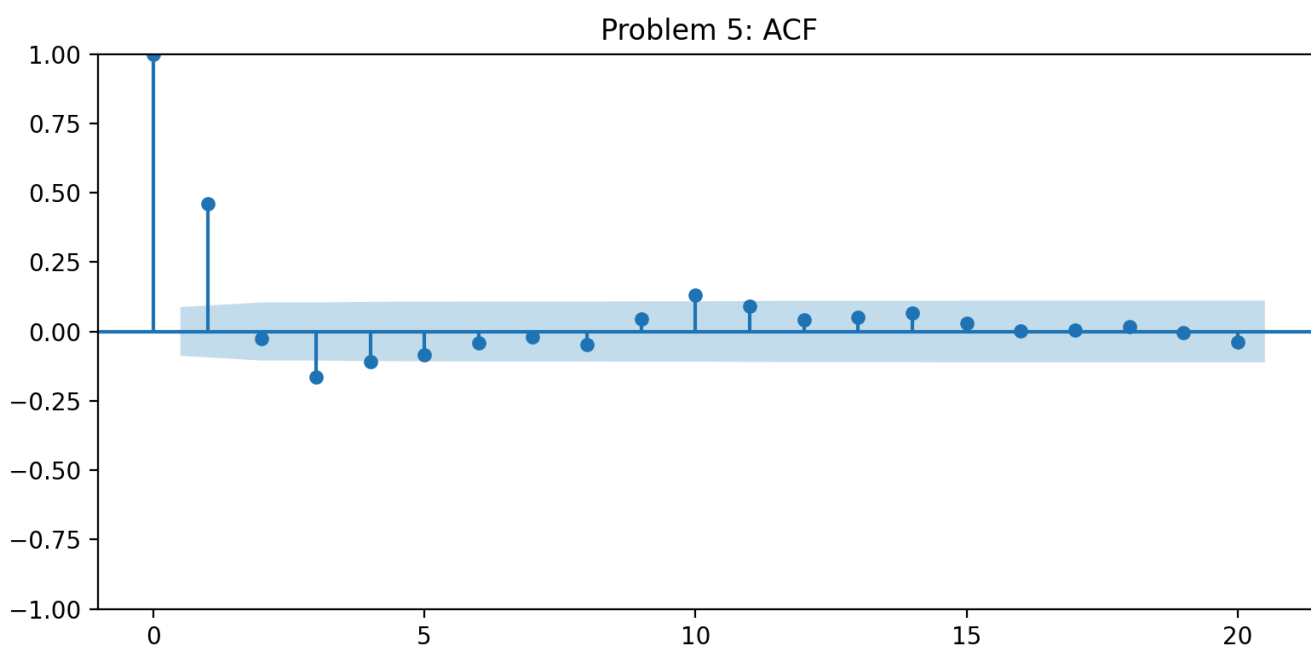
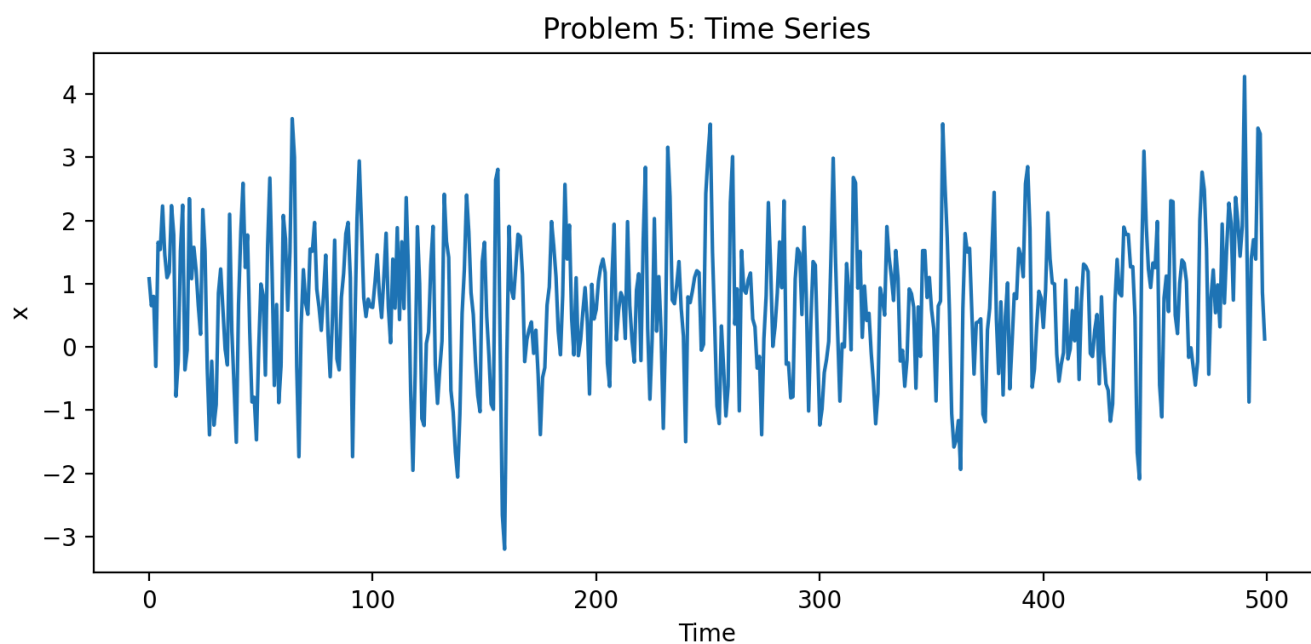
Near the center, the 95% band works very well. Farther away from the mean, however, more observations fall outside the band.

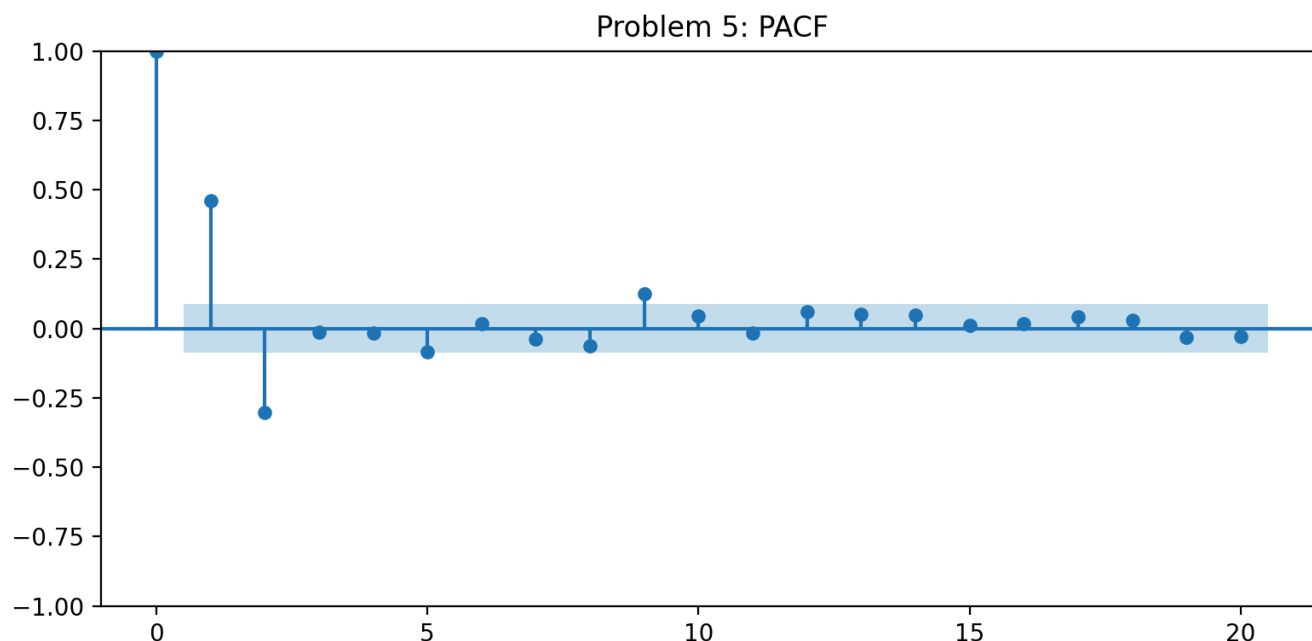
This suggests that the conditional variance is not actually constant. Therefore, the constant-variance assumption behind the multivariate Normal conditional distribution is not fully supported by the data.

The conditional mean can still be useful, and the coefficient Σ_{21}/Σ_{11} still corresponds to the OLS slope. The main part that fails is the constant-width uncertainty band.

5. Identifying an AR or MA Order

Predict





The time series appears stationary around a stable mean.

The approximate 95% significance band for the ACF and PACF is

$$\pm \frac{1.96}{\sqrt{500}} = \pm 0.0877.$$

The PACF has clear significant values at lags 1 and 2, followed mostly by values inside the significance band.

The ACF instead gradually decays with some oscillation.

Based on the rule from class, I predict an **AR(2)** process.

For an AR(p) process, the PACF should cut off after lag p , while the ACF should decay. For an MA(q) process, the opposite occurs.

Fit and Reconcile

The fitted AICc values are:

Model	AICc
AR(1)	1418.587
AR(2)	1372.438
AR(3)	1374.343
MA(1)	1389.084
MA(2)	1381.232
MA(3)	1374.117

The lowest AICc is for AR(2), so the fitted result agrees with my prediction.

For AR(2), the coefficients are approximately

$$\phi_1 = 0.6012$$

$$\phi_2 = -0.3026.$$

For AR(3), the third coefficient is only

$$\phi_3 = -0.0165.$$

This third coefficient adds very little to the model.

AICc rewards better fit but penalizes extra parameters. The small improvement from adding the third lag is not enough to offset the penalty for the extra coefficient, so AICc selects AR(2).

Ordinary R^2 would not make the same choice because adding another variable cannot reduce R^2 . Even an almost useless extra lag can slightly increase it. This is why AICc is more useful for choosing between models of different sizes.